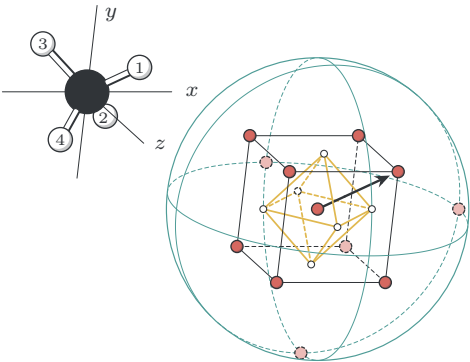
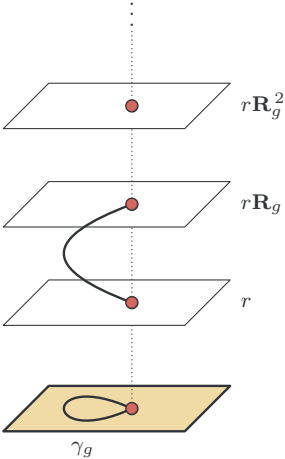


methane, CH₄, the reference **X**₀ in its body frame
 its twofold axes *x*, *y*, *z* pass through midpoints of opposite edges



the orientation space *SO*(3) as a ball
 direction gives the axis and distance the angle
 the twelve rotations of *T* in red
 the cell *U* at *q* = 0 in gold
 the arrow lifts the third-turn about bond 1
 from *r* to *rR*_{*g*}



over the base *SO*(3)/*T* lie twelve sheets
 one for each rotation in *T*, three drawn
 the third-turn's path γ_g is a loop in the base
 lifted, it climbs from sheet *r* to sheet *rR*_{*g*}
 A sees the climb as 1, ¹E as $e^{2\pi i/3}$
²E as $e^{4\pi i/3}$ and T as *x* → *y* → *z*